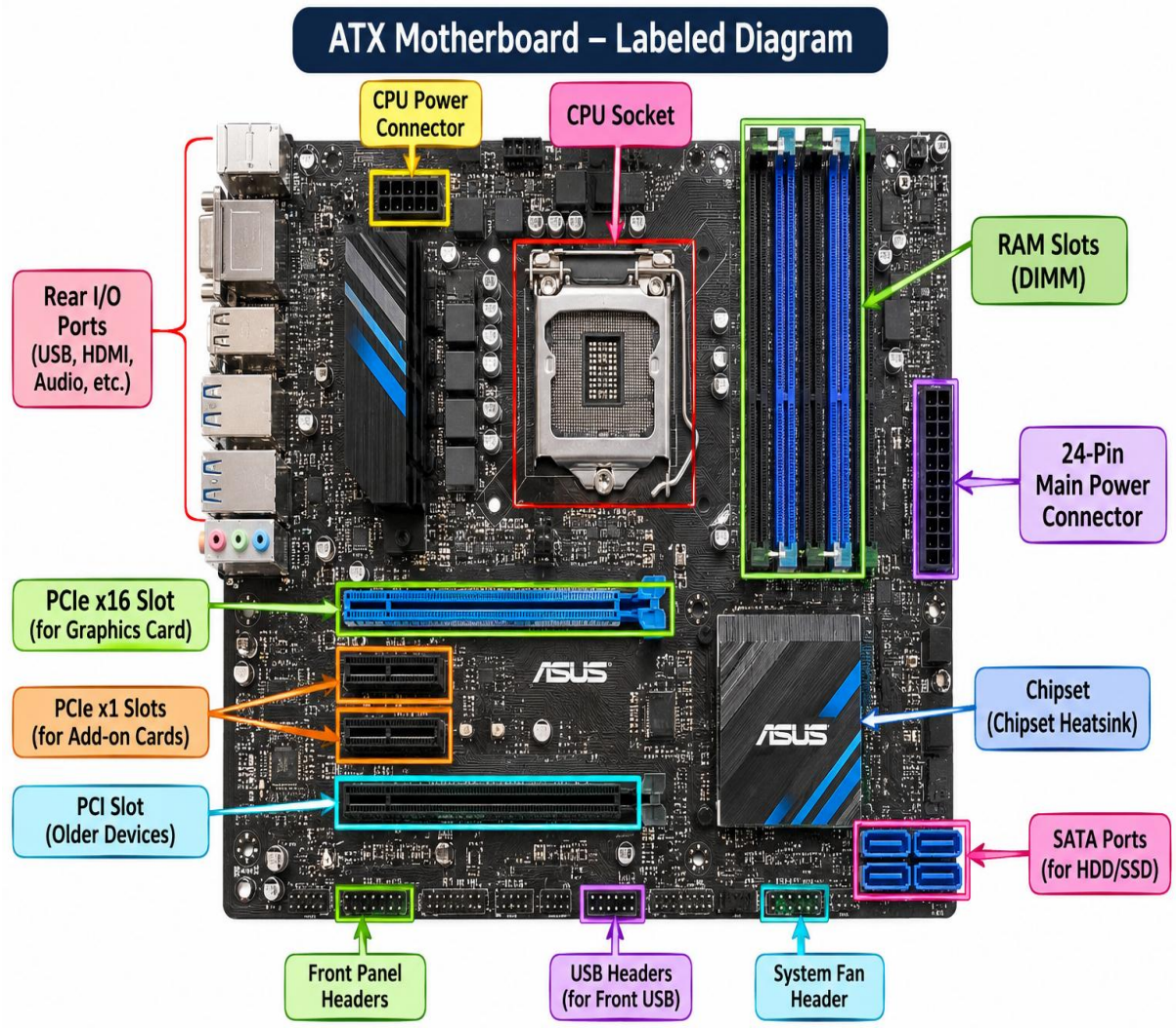
Motherboard and CPU Theory

1.Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2.Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type	Compatible CPU Model
LGA1151	Intel Core i7-9700
AM4	AMD Ryzen 5 5600
LGA1200	Intel Core i7-10700

1. **LGA1151**

- Developed by Intel.
- Supports selected 6th, 7th, 8th and 9th Generation Intel processors.
- Example: Intel Core i7-9700.

2. AM4

- Developed by AMD.
- Supports several generations of AMD Ryzen processors.
- Example: AMD Ryzen 5 5600.

3. LGA1200

- Developed by Intel.
- Supports selected 10th and 11th Generation Intel processors.
- Example: Intel Core i7-10700

3.Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

The CPU (Central Processing Unit) is the brain of the computer. It processes instructions, performs calculations, and controls the activities of the computer. It helps applications run smoothly.

The chipset is a group of electronic components on the motherboard that manages communication between the CPU and other hardware components, such as RAM, storage devices, USB ports, and other connected devices.

How they work together when opening Spotify:

1. When we open Spotify, the CPU processes the instructions needed to start the application.
2. The chipset helps transfer data between the CPU, RAM, and storage.
3. The CPU uses RAM to access the data needed to run Spotify.
4. The CPU processes the music playback instructions.
5. The chipset helps manage communication with the audio hardware so that we can hear the music through speakers or headphones.

4.Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

